VirtCraft

KubeVirt VM Management CLI

Craft, deploy, and manage KubeVirt virtual machines with a powerful Rust CLI. 44 OS templates, 8 resource profiles, multi-VM blueprints, interactive TUI, and full lifecycle management.

Rust-Powered — Single Binary — Production Ready

Why VirtCraft?

The kubectl alternative purpose-built for KubeVirt VMs.

Purpose-Built for KubeVirt

Not a generic K8s tool. Every command is designed for VM lifecycle: create, start, stop, clone, snapshot, export, health-check. No YAML required.

44 OS Templates

Ubuntu, Fedora, CentOS, Debian, RHEL, AlmaLinux, Rocky, Alpine, Arch, Windows 10/11/2022, FreeBSD, Talos, Flatcar, and more. One command to deploy.

Interactive TUI Dashboard

Rich terminal UI with real-time VM stats, resource gauges, activity feed, sparkline charts, and keyboard navigation. No browser needed.

Rust Performance

Single binary, zero dependencies, instant startup. Compiled with LTO for minimal size. Shell completions for bash, zsh, fish, PowerShell.

44	8	60+	11MB
OS Templates	Resource Profiles	CLI Commands	Binary Size

Feature Overview

Complete KubeVirt VM management from the command line.

VM Lifecycle

Command	Description
virtcraft create	Create a VM from template, profile, or YAML config
virtcraft list	List all VMs with status, CPU, memory, IP, node, age
virtcraft start / stop / restart	Control VM power state via KubeVirt RunStrategy
virtcraft status	Detailed VM status with resources, volumes, conditions
virtcraft clone	Clone an existing VM with new name
virtcraft export	Export VM configuration to YAML/JSON
virtcraft delete	Delete a VM with confirmation
virtcraft generate	Generate VM manifest without creating it
virtcraft validate	Validate a VM configuration file before deployment
virtcraft diff	Compare two VM configurations side by side

Quick Start

```
# Create an Ubuntu 24.04 VM with 4 CPUs and 8GB RAM
$ virtcraft create --name web-01 --template ubuntu-24.04 \
    --cpus 4 --memory 8Gi --disk 50Gi

# List all VMs
$ virtcraft list
NAME          NAMESPACE   STATUS    RUNNING   CPU       MEMORY
```

```
web-01        default     Running   Yes       4 cores   8Gi

# Take a snapshot
$ virtcraft snapshot create web-01 --name before-upgrade

# Use the interactive TUI
$ virtcraft tui
```

Templates & Profiles

Pre-configured OS templates and resource profiles for every workload.

OS Templates (44 total)

Red Hat Family

- RHEL 8, 9
- CentOS Stream 8, 9
- Fedora 39, 40, 41
- AlmaLinux 8, 9
- Rocky 8, 9
- Oracle Linux 8, 9

Debian Family

- Ubuntu 24.04, 22.04, 20.04, 18.04
- Debian 11, 12

SUSE Family

- openSUSE Leap, Tumbleweed

Windows

- Windows 11, 10
- Windows Server 2022, 2019

Others

- Alpine, Arch Linux
- FreeBSD 13, 14
- Flatcar, Talos

Resource Profiles (8 built-in)

Profile	CPU	Memory	Disk	Use Case
dev	1 core	1 Gi	10 Gi	Development & testing
small	2 cores	2 Gi	20 Gi	Light workloads
medium	4 cores	4 Gi	40 Gi	General purpose
large	8 cores	16 Gi	100 Gi	Production services
database	4 cores	16 Gi	200 Gi	Database servers
web	2 cores	4 Gi	40 Gi	Web/API servers

high-perf	16 cores	64 Gi	500 Gi	HPC, analytics
prod	8 cores	32 Gi	200 Gi	Production critical

Blueprints & Advanced

Multi-VM stacks, snapshots, health checks, and API server.

Multi-VM Blueprints

Deploy complete application stacks with a single command. Blueprints define multiple VMs with dependencies, networking, and resource profiles.

LAMP Stack

Linux + Apache + MySQL + PHP. 3 VMs with automatic dependency ordering.

Kubernetes Cluster

1 control plane + 2 worker nodes. Pre-configured for kubeadm bootstrap.

3-Tier Architecture

Web frontend + application server + database. Network isolation.

CI/CD Pipeline

Jenkins/GitLab + build agents + artifact storage. Ready to use.

Snapshot Management

```
$ virtcraft snapshot create web-01 --name before-upgrade
$ virtcraft snapshot list web-01
$ virtcraft snapshot restore web-01 --name before-upgrade
$ virtcraft snapshot delete web-01 --name before-upgrade
```

Health Checks & Recommendations

```
$ virtcraft health-check
  System Health Score: 87/100
  [OK]  CPU utilization normal (45%)
  [OK]  Memory utilization normal (62%)
```

```
  [WARN] Disk usage above 75% on 2 VMs
  Recommendation: Consider expanding disk for database-01
```

REST API Server

```
$ virtcraft api-serve --port 8080
$ curl http://localhost:8080/api/v1/vms
$ virtcraft api-routes    # List all endpoints
$ virtcraft api-spec      # Generate OpenAPI spec
```

Interactive TUI

Full-featured terminal dashboard for VM management.

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Dashboard View

Real-time cluster overview with VM counts, CPU/memory/disk gauges, sparkline charts for trends, recent activity feed, and quick navigation.

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VM List View

Sortable table with name, status, CPU, memory, IP, node, age. Search and filter. Multi-select for batch operations. Keyboard navigation.

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VM Details View

Comprehensive VM info: status, conditions, resources, volumes, network. Start/stop/restart actions directly from the TUI.

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Snapshot View

List, create, and manage snapshots. Ready/pending status. One-key snapshot creation.

Multi-Tenant RBAC

Command	Description
virtcraft tenants-create	Create isolated tenant with namespace and quotas
virtcraft users-create	Create user with role assignment
virtcraft roles-create	Define custom RBAC roles
virtcraft quotas-create	Set resource quotas per tenant

virtcraft groups-create	Organize users into groups

Additional Features

Webhooks

Register webhooks for VM events: created, started, stopped, failed, snapshot, migration.

Config Management

Save, load, list, delete reusable VM configs. Template library for team sharing.

Shell Completions

Auto-generated for bash, zsh, fish, PowerShell, elvish. Tab-complete everything.

Technology Stack

Built with Rust for performance, safety, and reliability.

Component	Technology	Details
Language	Rust	Memory-safe, zero-cost abstractions, single binary
K8s Client	kube-rs	Native async Kubernetes client
CLI Framework	clap v4	Derive-based args with shell completions
TUI Framework	ratatui	Terminal UI with charts, gauges, tables
Terminal	crossterm	Cross-platform terminal control
Serialization	serde	YAML, JSON, TOML configuration
Build	LTO + strip	11MB optimized binary
Tests	54 unit + 8 doc	Comprehensive test coverage

Integration with hyper2kvm Ecosystem

hypersdk (export) → hyper2kvm (migrate) → VirtCraft (CLI manage) → v9s (web manage)

VirtCraft — KubeVirt management, done right

44 OS templates. 8 profiles. Multi-VM blueprints. Interactive TUI.
One binary. Zero dependencies. Instant productivity.

VirtCraft — KubeVirt VM Management CLI
Built with Rust — Proprietary Software
Founders: Susant Sahani & Sibu